

An Ubiquitous Two-factor Authentication (2FA) Platform for IoT Applications

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Abstract—IoT-enabled smart wearables have become ubiquitous, offering continuous sensing and real-time data exchange through embedded sensors. Utilizing these capabilities, we propose a robust two-factor authentication (2FA) system that combines fingerprint and ear-canal-based biometrics. The fingerprint modality uses a capacitive sensor to address challenges posed by wet or oily fingers, while the ear-canal modality exploits the unique anatomical and acoustic properties of the ear-canal captured using standard earphones with built-in microphones. The system was evaluated with 30 participants under varied usage conditions. An inertial measurement unit (IMU) sensor placed near the earphone established a reference orientation during enrollment. Identity matching performed using the sum of squared distances (SSD) achieved a 100% recall for earphone orientation changes up to $\pm 25^\circ$. Our system recorded a best area under the curve (AUC) of 99.68% and an equal error rate (EER) of 0.7% for authentication. Offset and drift correction were applied to the IMU data to ensure stable and reliable performance over time, which resulted in a recall improvement from 97% to 98.4% over one month, highlighting the system's robustness.

Index Terms—authentication, biometrics, fingerprint, ear-canal, acoustic, human-centered computing, and ubiquitous

I. INTRODUCTION AND MOTIVATION

Authentication is fundamental to safeguarding system security, privacy, integrity, and trust. It ensures that only authorized individuals can perform designated actions. Among the various authentication mechanisms, fingerprint recognition is one of the most widely employed methods [1], [2], typically relying on optical sensors to capture signatures. However, the effectiveness of fingerprint-based systems is often challenged by factors such as harsh environmental conditions, aging, physical disturbances, inconsistencies in lighting, and environmental changes [3]. A significant limitation of the commonly used optical sensors is their inability to effectively capture data from wet or oily fingers. Our system replaces the traditional optical sensor with a capacitive sensor, enabling reliable image capture and authentication even for wet or oily fingers.

To improve system robustness, multifactor authentication is commonly utilized [4], combining approaches like passwords, physical security tokens, and a range of biometric techniques such as fingerprint recognition [2], facial identification [5], ear-canal shape-based authentication, and voice-based authentication [6]. Beyond traditional biometrics, bioelectrical signals, such as electrocardiograms (ECG) and electroencephalograms

(EEG), are used as alternative means of identification [7]. These factors motivated us to design a two-factor authentication (2FA) device that leverages distinctive biometric signatures.

Ear-based authentication is a relatively new technique and is gaining traction, especially in scenarios requiring passive identification [2]. This method analyzes the unique anatomical features of the outer ear, such as the auricle or pinna, including structures like cartilage, tragus, helix, and lobule, which are distinct to each individual [8]. The pinna plays a crucial role in directing sound waves into the ear-canal, where eardrum vibrations initiate the auditory process. The length of the ear canal significantly impacts its resonant frequency. In general, a shorter ear-canal length results in a higher resonance frequency, while a longer length leads to a lower resonance frequency.

In this work, we design and build an ubiquitous 2FA device that combines fingerprint recognition with ear-canal shape-based authentication. This offers a novel approach to secure and reliable identity verification. The following are the contributions in this work:

- Reliable and robust hardware for ear authentication has been developed. This new hardware is a significant improvement over previous methods because it can record the ear and head orientations during the authentication and reduces false rejections of genuine users caused due to positional shifts.
- An empirical model of the ear authentication is proposed from sensor data.
- An ubiquitous electronic product prototype is constructed for the purposes of 2FA.

II. RELATED WORK

Ear recognition provides greater stability compared to facial recognition, which can be influenced by changes in expression [9], light variations, misalignment in the face, or variation in poses [10]. Facial recognition is also not a solution where face masks are worn continuously. Previous studies have explored ear images for biometric authentication [11], but challenges such as illumination variations and occlusion persist. Bilateral ear acoustics, as tested in [12], utilize error signals derived from the differences between signals from both ears. Research in [13] investigates in-ear authentication in virtual reality

systems, where users perceive 3D sounds through binaural earphones placed in different positions. Other studies, such as [14], and [15], analyze the variations in acoustics due to the external ear's shape but do not employ modeling techniques specifically for authentication.

Efforts in human authentication using ear acoustics are detailed in works such as [7], [16], and [17], which utilize acoustic signals for individual identification. The advent of miniature microphones and loudspeakers with flat frequency responses has further fueled interest in ear-based authentication systems. Work in [17] explores acoustic ear recognition using probes that transmit sound signals and record echoes, requiring no active participation from the subject, albeit using audible signals. Holistic developments in acoustic ear recognition are discussed in [18] to achieve a better error rate.

A concept for silent authentication is introduced in [19], enabling repetitive and continuous verification. Meanwhile, [20] presents a general-purpose Arduino-based device designed for open sensing, equipped with a range of sensors and actuators for research and development purposes. These developments emphasize the increasing significance of ear-based biometrics in enhancing the capabilities of multimodal authentication systems.

III. UBIQUITOUS TWO-FACTOR AUTHENTICATION (2FA) SYSTEM

A. Features

Our ubiquitous 2FA system is designed considering the following features.

Ubiquitous: Wired and wireless earphones are widely used due to their availability and user familiarity. We develop a custom-designed PCB add-on with a form factor of 22×19 mm as shown in Fig. 1. This low-cost module interfaces directly with the earphones, minimizing additional hardware requirements and reducing the overall system cost.

Unique: Recent studies have identified the ear as a viable

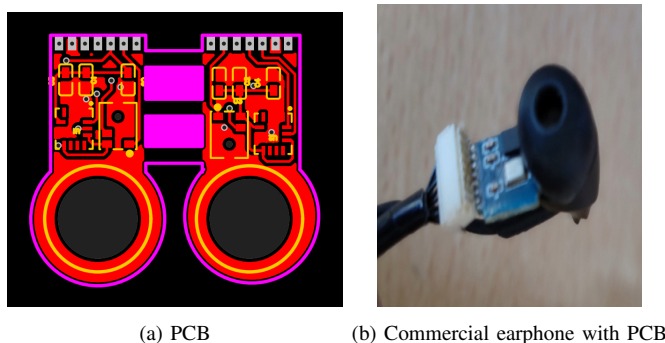


Fig. 1: Ear-canal shape-based authentication module PCB

passive biometric modality [8]. Compared to facial features, the ear exhibits greater stability, as it is less susceptible to variations caused by emotional expressions.

Robust: Robustness is a fundamental design requirement for

any reliable biometric authentication system. The signature obtained from the ear remains stable despite earphone misalignment caused by motion artifacts. The capacitive fingerprint sensor-based authentication further hardens the robustness.

Repeatability: The system demonstrates a high probability of successful authentication across various operating conditions.

Not obtrusive: A hassle-free enrollment and authentication system with no active involvement from the user.

Fig. 2 shows the fingerprint and ear authentication modules. The process involves two steps: enrollment and authentication. During the enrollment phase, participants first have to register their fingerprints, followed by ear-canal shape-based data collection.

B. Working of the fingerprint-based authentication

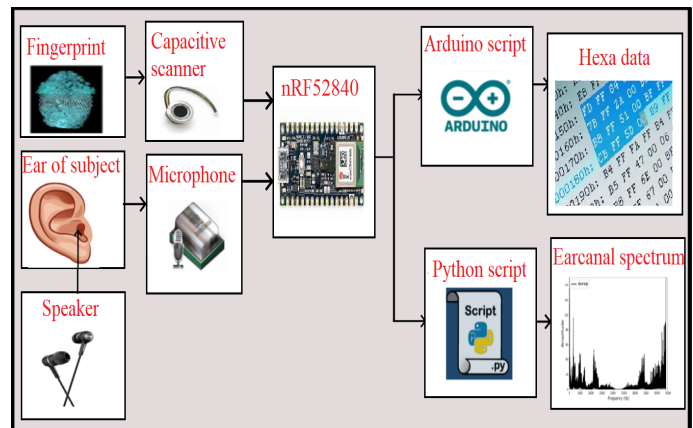


Fig. 2: Functional block diagram of our 2FA IoT system

Fingerprint data is captured using a capacitive sensor, which consists of a grid of tiny capacitors that store electrical charge. When fingerprint ridges make contact with the sensor, the charge is discharged at those specific points. The sensor utilizes human conductivity to create an electrostatic field, which is then processed into a digital image. Each signature data is of 132 bytes. This sensor module has a fast scanning time of less than 0.2 seconds and a high-security false rejection rate (FRR) of $\leq 1\%$. The sensor module does the image rendering, calculation, feature finding, and searching.

The fingerprint data is acquired by the microcontroller through a UART (universal asynchronous receiver-transmitter) interface. The microcontroller processes the data and saves the corresponding hexadecimal bit representation.

C. Working of ear-canal shape-based authentication

Fig. 2 illustrates the functional flow of the acoustic ear-canal shape-based authentication method. An acoustic signal is sent to the ear-canal using standard commercial earphones, and a microphone is placed close to the earphones to capture the reflected echo signals, which are further processed. For our analysis, we generated an 18-second-long audio signal in the frequency range of 1 Hz - 22 KHz at 53 dB, covering the entire audible frequency range. The audio/sound signal is generated

using an open-source online tool. A microphone positioned near the earphones captures the reflected echo signals. These captured signals are then sent to the microcontroller for processing. This approach, commonly referred to as earable authentication [21], uses the distinct acoustic properties of the ear-canal for identification. To enhance robustness, the system includes a 6-axis inertial measurement unit (IMU) sensor colocated with the microphone. The IMU sensor measures the precise angle of the earphone relative to the ear during data collection. This ensures accurate orientation and minimizes errors caused by microphone positioning variations, a limitation often noted in prior studies. By addressing these challenges, our system improves the reliability of the ear-canal shape-based biometric authentication process.

IV. DESIGN AND ELECTRONICS

In this section, the microcontroller unit, sensors used in our system, and software functions are detailed. Fig. 3 illustrates the key hardware components of our system.

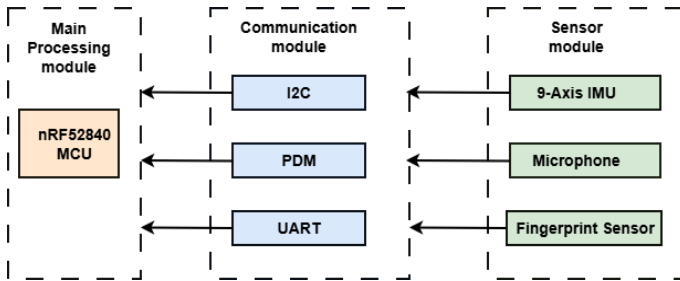


Fig. 3: Hardware modules of the 2FA IoT system

A. Main processing module

The central processing unit of our system is the nRF52840 system-on-chip (SoC) featuring the NINA-B306-00B module. The clock frequency is at 32 MHz. The nRF52840 is built on an ARM Cortex-M4 core and includes Bluetooth 5 support for wireless communication. It supports UART, I2C, and SPI communication protocols, enabling seamless data acquisition from connected sensors. Additionally, the nRF52840 is compatible with the Arduino IDE, making it versatile for development.

B. Sensor module

Our system incorporates several sensors; the details of key sensors are discussed below.

1) *Fingerprint sensor*: The system employs the Adafruit R503 capacitive fingerprint sensor, which operates using the UART protocol and is compatible with the Arduino IDE. This sensor was selected for its rugged panel-mount design, ease of integration, and high environmental resilience. The module supports both one-to-one and one-to-many matching methods.

2) *Microphone*: We use the SPH0641LU4H-1 acoustic ultrasonic microphone, co-located near the speaker. The SPH0641LU4H-1 is a compact, high-performance, and low-power digital microphone with a single-bit pulse density modulation (PDM) output. The sensor operates in a voltage range of 1.62 V to 3.6 V (standard performance mode), featuring a

sensitivity of -26 dBFS and an operating current of 1 mA. This microphone offers a high signal-to-noise ratio (SNR) of 64.3 dB and an omnidirectional pickup pattern, ensuring effective sound capture for our system analysis. The sensor module integrates an acoustic sensor, a low-noise input buffer, and a sigma-delta modulator. Being a PDM microphone, it employs pulse density modulation to convert acoustic signals into digital data, making it suitable for our audio analysis.

3) *IMU sensor*: The system integrates a BMX160 low-power IMU sensor, compatible with both I2C and SPI communication protocols. The sensor was selected for its ultra-compact form factor, energy efficiency, integrated 9-axis motion sensing, and robust motion detection capabilities. This integrated sensor delivers accurate measurements of acceleration, angular rate (gyroscopic), and geomagnetic fields in all spatial directions. It is used to calculate roll and pitch angles during ear authentication.

4) *Earphones*: Our system uses the standard commercial earphones available in consumer electronic stores shown in Fig. 1b. For this implementation, we utilize standard wired earphones featuring 10 mm dynamic drivers and a resistance of 16 Ohms. The earphones provide a sensitivity of 92 dB with a frequency response of 20 Hz - 20 KHz supporting the full human audible frequency range. The earphones have silicone earbud tips and a 3.5 mm jack cable adapter to connect to a laptop or a desktop. Our system is compatible with any standard wired or wireless earphones that meet the above-mentioned technical specifications.

C. Communication module

Efficient communication protocols are essential for seamless sensor integration and data acquisition. The choice of protocols depends on the compatibility of the sensor modules with the microcontroller unit (MCU).

1) *UART*: UART is a serial communication protocol used in the system to interface the fingerprint sensor with the MCU over a 57600 bps baud rate.

2) *I2C*: The inter-integrated circuit (I2C) protocol facilitates serial communication using only two bidirectional open-drain lines: SDA (serial data line) and SCL (serial clock line). Both lines are pulled high during operation. In our system, the IMU sensor is connected to the microcontroller via the I2C protocol and configured to run at 400 kHz.

3) *SPI*: The serial peripheral interface (SPI) is a synchronous serial communication protocol known for its high-speed data transfer capabilities. It uses four signal lines: MISO (master in slave out), MOSI (master out slave in), SCLK (serial clock), and CS (chip select). An SD card is interfaced to the MCU using SPI communication configured to run at 8 MHz.

D. Software design

The software facilitates user interaction with the hardware. It encompasses two primary functions: data acquisition and data processing. Data acquisition is associated with the enrollment of an individual, while data processing authenticates the individual.

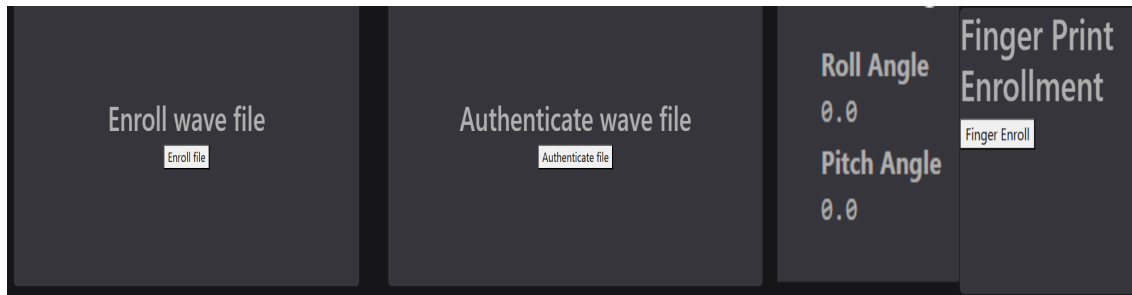


Fig. 4: Dashboard of our 2FA IoT system

1) *Data acquisition for enrollment:* Data from fingerprint and acoustic sensors are captured using a graphical user interface (GUI) illustrated in Fig. 4. The fingerprint sensor captures data from each individual, which is then processed into a 768-byte template and stored with a unique ID. For ear acoustic data collection, the system is Bluetooth-enabled, and the required IMU sensors are selected. The audio signal sampling rate is set at 44,100 Hz to ensure high-quality data acquisition. The system captures the reflected/echo signal from a pre-recorded audio file and stores it with an ID. Slight movements of the head, mouth, and face were allowed to mimic realistic conditions. The roll and pitch angles from the IMU sensor are continuously displayed on the GUI, enabling real-time tracking of the microphone and speaker orientation relative to the ear.

During the enrollment process, echo signals (saved as `<name>.wav`) are captured and processed to extract unique biometric features using the following steps:

- **MFCC extraction:** The system reads a `.wav` audio file (echo signal), performs FFT, and extracts mel-frequency cepstral coefficients (MFCCs), which characterize the acoustic features. The MFCCs (c_n) are computed using the relation in (1).

$$c_n = \sum_{m=1}^M \log S(m) \cos \left(\frac{\pi n}{M} \left(m - \frac{1}{2} \right) \right) \quad (1)$$

Where $n = 1, \dots, N$ is the index of cepstral coefficients, and N is the length of the audio files, $S(m)$, $m = 1, 2, \dots, M$, represents the output of an M -channel filter bank.

- **Dimensionality reduction:** Principal component analysis (PCA) is applied to the averaged MFCCs to reduce feature dimensions while retaining important patterns.
- **Clustering:** The PCA-reduced data is clustered using the K-Means algorithm ($K = 7$), forming a biometric signature for each user.

The enrollment process continuously monitors a designated directory for incoming `.wav` files. Upon acquiring three recordings (for each individual), MFCC features are extracted. These features are then averaged and processed using PCA followed by K-means clustering. The transformed data and trained model are stored, this ends the enrollment phase.

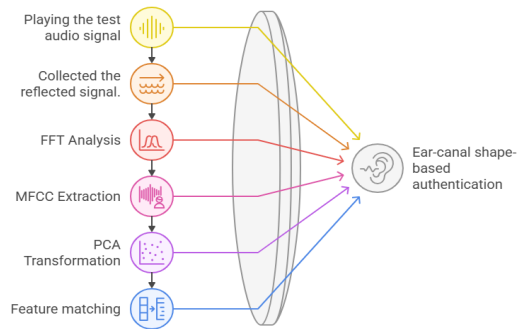


Fig. 5: Data processing flow for ear authentication

2) *Data processing for authentication:* The processing phase plays a vital role in analyzing and extracting meaningful insights from the acquired data. For fingerprint authentication, a 1:N template matching method is applied.

For each ear-canal shape-based authentication, a pre-recorded audio clip is transmitted through the ear-canal, and the echo signal is recorded. To facilitate matching, acoustic features are extracted from the authentication audio files. The matching phase compares echo recordings against enrolled data to authenticate individuals. We implemented a hybrid strategy for user access. First, we perform 1:N identification to determine the user's identity from a database of 'N' enrolled subjects. It is followed up with a more stringent 1:1 authentication to verify that the user is indeed who they claim to be. The workflow of feature extraction is illustrated in Fig. 5.

- **Feature extraction:** Similar to enrollment, MFCCs are extracted from the new `.wav` file, followed by PCA transformation.
- **Data retrieval:** For each enrolled user, PCA features and K-Means models are applied.
- **We use 3 matching techniques:**
 - **Sum of squared distances (SSD):** The distance of the new sample from cluster centroids is calculated. The user with the smallest sum is selected.
 - **Cosine similarity (CS):** Measures the angular similarity between the enrolled and new sample vectors.
 - **Euclidean distance (ED):** Calculates straight-line distance in feature space between the new sample and enrolled users.

The matching process continuously monitors for new audio files. Upon detection, it applies all three matching techniques and prints the three best matches for each.

V. EVALUATION AND RESULTS

A. Data collection

The ethical clearance for the study was obtained from the university institute committee (IHEC). This committee ensures that all research involving human subjects is in accordance with strict guidelines to protect the well-being and privacy of the participants. Informed consent was obtained from all participants prior to data collection and carried out with necessary safety precautions.

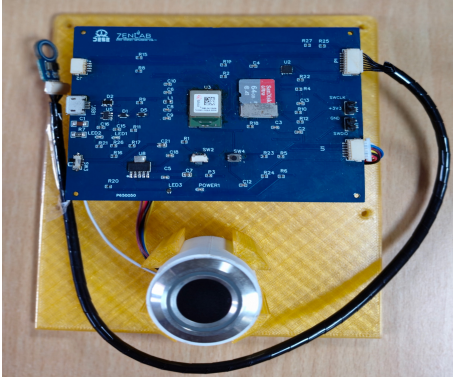


Fig. 6: Prototype of our 2FA IoT system

To evaluate the robustness and efficacy of the proposed system, multimodal biometric data comprising fingerprint and ear-canal shape-based acoustic signals was collected from 30 individuals across genders, aged between 20 and 65 years. For the acoustic modality, recordings were obtained from the left and right ears from 21 subjects. All participants had normal hearing capabilities, and the data collection was carried out in both a quiet environment and a typical laboratory setting. The ambient noise levels ranged between 40 dB and 55 dB. For fingerprint data acquisition, users may enroll using any finger of their choice. Fig. 6 shows our prototype system.

B. Resonance spectrum

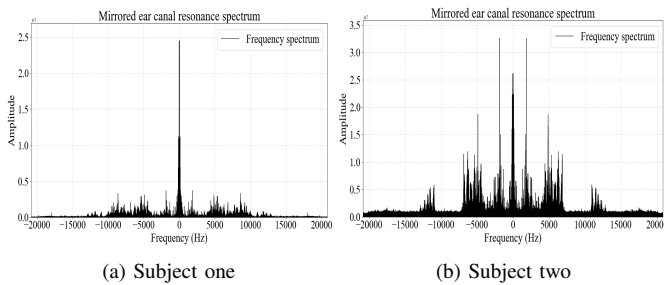


Fig. 7: Ear-canal resonance spectrum of two subjects

Fig. 7a and Fig. 7b illustrate the spectral peaks representing the resonance frequencies captured from the left ears of subjects one and two by using the same test audio signal. The

resonant frequency's peaks are shown in Fig 7, the peak height reflects the resonance strength, and the peak width signifies the resonance attenuation. It is evident from the figures that the resonant frequencies, along with the height and width of the peaks, are distinct for each individual, highlighting their uniqueness.

C. System evaluation

We evaluate our system by using a hybrid strategy. First we perform identification in which a subject is matched using the 1:N matching technique. One subject is matched against all the subjects enrolled in the dataset. We find 3 closest matches to the subject using SSD, CS, and ED methods explained in Section IV-D2. After identifying a subject from the enrolled database using 1:N matching, the system proceeds to the authentication stage. We build a binary support vector machine (SVM) training dataset to distinguish between genuine users and impostors. The impostor subset is composed of random 10% sample of all users, as well as the two best-matching individuals from the preceding identification step. We compute decision scores and apply thresholds to calculate false acceptance rate (FAR) and false rejection rate (FRR). These determine the equal error rate (EER) and area under the curve (AUC) metrics for our system. In summary, just relying on the 1:N matching technique results in a higher false acceptance ratio, especially when the dataset is large. Combining 1:1 (SVM) with 1:N matching improves security and leads to strong verification. Thus, this method reduces the overall EER, FAR, and AUC.

FAR is the case where an imposter is wrongly accepted as a genuine subject. FRR is the rate at which a genuine subject is wrongly rejected. True acceptance rate (TAR) is the rate at which the genuine subject is correctly accepted ($TAR = 1 - FRR$).

$$FAR = \left(\frac{T_{FA}}{T_{auth}} \right) \times 100\% \quad (2)$$

$$FRR = \left(\frac{T_{FR}}{T_{auth}} \right) \times 100\% \quad (3)$$

Where, T_{FA} is the number of imposters wrongly accepted, T_{FR} is the number of genuine subjects wrongly rejected, and T_{auth} is the total number of authentications performed. For the classification model, the threshold trade-off determines the system's performance for matching probability. A low threshold makes it easier to accept subjects (low FRR) but also easier to mistakenly accept impostors (high FAR). A high threshold makes the system more secure against impostors (low FAR) but more likely to reject genuine subjects (high FRR). Hence, there is a trade-off between FAR and FRR. The detection error tradeoff (DET) curve indicates the relationship between FAR and FRR, in which the intersection between the FAR and FRR is known as the EER, which is used as a performance indicator of biometric authentication.

The receiver operating characteristic (ROC) curve is used to plot the relationship between TAR and FAR. The AUC measures the overall performance across all thresholds. An

AUC of 1.0 represents perfect accuracy, while an AUC of 0.5 indicates performance no better than random chance. A higher AUC signifies a more accurate system. We computed the EER and AUC for each of the 51 datasets and evaluated the mean. The best, worst, and mean AUC and EER values are tabulated in Table I.

TABLE I: AUC and EER evaluation of the system for ear-canal shape-based authentication

	AUC	EER
Mean	98.3%	0.7%
Best value	99.68%	0.05%
Worst value	96.9%	0.9%

From Table I, it is evident that our system demonstrates a high level of accuracy for user authentication, as evidenced by a mean AUC value of 98.3%, and a very low EER.

We assessed system performance using F-score, precision, and recall. Given the true positive (TP), false negative (FN), and false positive (FP), the precision and recall for the system are defined in (4) and (5).

$$Precision = \frac{TP}{TP + FP} \quad (4)$$

$$Recall = \frac{TP}{TP + FN} \quad (5)$$

where a high precision means that only the authorized subjects/users pass the verification, and a high recall indicates that genuine users will not be rejected. F-score, is defined in (6).

$$F\text{-score} = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (6)$$

The F-score combines the concepts of precision and recall into one score, providing a more balanced view of performance, especially for a large imbalanced dataset.

D. Setting the baseline time interval

To optimize authentication speed without sacrificing accuracy, it is essential to determine the minimum required number of time samples. As shown in Fig. 8, accuracy improves with increasing time window size, starting from 1 second. Although performance metrics continue to rise with longer windows, they plateau around 5 seconds, leading us to select a 5-second window as the optimal balance between speed and accuracy. In total, this evaluation per subject generated approximately 1.2 million data points by taking data from a 5-second window.

TABLE II: Ear-canal shape-based authentication accuracy of the genuine and imposters with 5 second time window

	Mean	Median	Standard deviation
Precision (%)	99.73	100	1.97
Recall (%)	99.68	100	1.30
F-score (%)	99.70	99.02	1.92

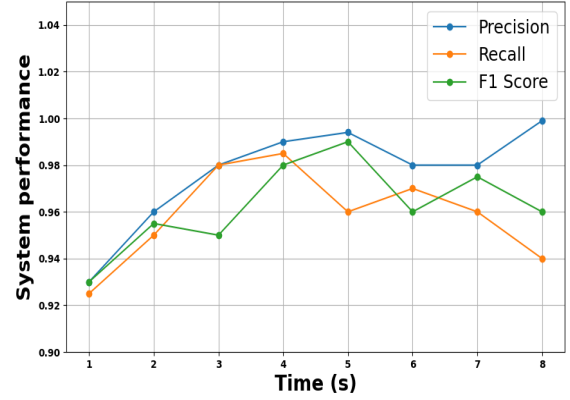


Fig. 8: System performance for different time windows

Table II summarizes the recall, precision, and F-score results for genuine users tested against 10% imposters using this 5-second window.

E. Performance of ear-canal shape-based authentication

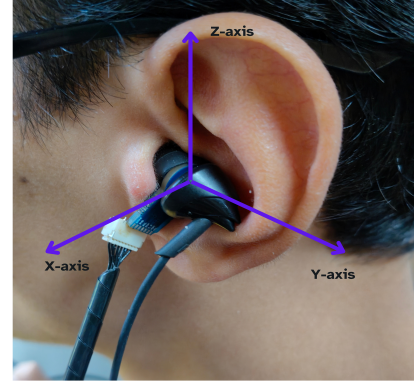


Fig. 9: Earphone position and the X, Y, and Z axes

The users were authenticated under two scenarios, i.e., authentication soon after enrollment and authentication after an extended period. In the initial case, generally the earphone is in the same position during enrollment and authentication. The earphone is not removed from the ear, this ensures that the relative position of the earphone with the ear does not change. Our authentication results are 100% soon after enrollment.

F. Earphone positions

The shape and size of each human ear is unique, and hence each individual has a tendency to wear the earphones suitable to match their comfort level, and thus there is no unique position to hold the earphones. Also, users place the earphones in different positions every time they wear the earphones, this changes the relative position of the earphone with the ear-canal. To study how earphone position affects our system, we

use an IMU to track the earphone's orientation relative to the ear. During the enrollment process, the IMU's X, Y, and Z values are noted, which are used as a reference orientation for each subject.

To assess the robustness of our system in terms of earphone positional variations, we evaluated its performance under a variety of earphone-wearing conditions, including position changes and intentional rotational deviations along all three axes. Fig. 9 shows the X, Y, and Z axes of the earphone. One week after initial enrollment, a subset of 18 participants (10 females and 8 males) was selected for this evaluation, where the earphone orientation was intentionally altered from the reference. Earphones were positioned at angles of 25° , 45° and 180° . Participants were asked to listen to the test audio clip under each modified earphone placement. Similar to the enrollment process, we emulated slight movements of the head, mouth, and face to mimic realistic conditions.

As shown in Table III, during enrollment, small rotational changes from the reference position of the earphone led to a minor decrease in authentication performance. As the rotation angle is increased beyond 45° , the acoustic properties of the reflected signal were significantly altered due to changes in the earphone-and-ear-canal interface, resulting in a drop in the matching score. The drop might be due to sound leakage and modified cavity volume. Among the matching methods, SSD demonstrated the most resilience to such changes. A near 100% authentication success is obtained when the earphone has a slack in orientation up to a maximum of $\pm 25^\circ$ from the reference. The reduction in recall varied based on the extent of earphone rotation. In summary, the built-in IMU sensor tracks earphone orientation, and the system maintains reliable authentication for positions within $\pm 25^\circ$. If the orientation exceeds this range, users are prompted to readjust the earphones, which reduces the rate of false rejections of genuine subjects caused by positional shift.

Studies in [22] and [23] indicate that the pinna (outer ear) continues to grow with age. Alongside this growth, the reduction in tissue elasticity can lead to a partial collapse of the cartilaginous ear canal. Such changes slightly reduce the canal volume and alter ear positioning, which in turn introduces variations in the captured echo signal. The captured echo signal is influenced by these anatomical changes. However, our system's ability to maintain reliable authentication within position variations of $\pm 25^\circ$ demonstrates sufficient robustness to account for age-related factors of the subject.

TABLE III: System performance under different earphone positions and recall values during authentication

Ear- phone position	Within $\pm 25^\circ$			Between 25° – 45°			more than $\pm 45^\circ$		
	SSD	CS	ED	SSD	CS	ED	SSD	CS	ED
Recall percent- age	100%	99%	97.4%	98.4%	96%	95.4%	90%	85.4%	84%

G. Time-based authentication evaluation

We conducted an accelerated test to gain insight into the ear authentication. Extensive fingerprint data was collected. Assuming users might wear the earphones at varying angles up to 180° during use, approximately seven samples per subject were collected for comparison with the enrolled pattern. Table IV shows the system's performance using the SSD matching method, with the second row (without applying offset and drift corrections) showing the results for over a one-month period post-enrollment. The evaluation involved five subjects.

Also, we observed drift and offset variations in the IMU readings when used for a prolonged period. To address this, we implemented offset and drift correction for the IMU sensor.

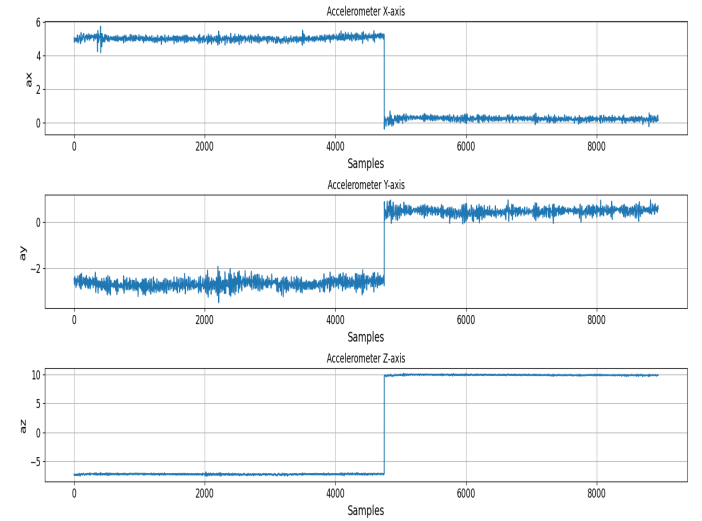


Fig. 10: Drift and offset correction of IMU sensor data

1) *Offset in IMU:* We observed that the IMU shows an initial bias in its readings upon power-up, known as turn-on bias stability. This arises due to sensor imperfections and environmental influences. The offset, which may vary with each startup, impacts both accelerometer and gyroscope data, resulting in measurement inaccuracies. Hence, for reliable data collection, accurately estimating and compensating for the bias at the beginning of every session becomes essential.

TABLE IV: System performance using SSD matching method for a period of one month with and without offset correction

Recall of SSD (in %)	1 day	1 week	1 month
Without offset and drift correction	100%	99.4%	97%
With offset and drift correction	100%	99.7%	98.4%

2) *Drift in IMU:* We observed a gradual drift in the sensor readings during prolonged operation. Drift refers to the slow deviation in a sensor's output over time, even when the actual measured parameter remains unchanged. Such variations can stem from sensor limitations, environmental changes, temperature shifts, or mechanical stress. This drift affects the IMU's

accuracy, which diminishes over time, leading to an increasing discrepancy between the measured and actual orientation.

The offset of the IMU was estimated, and for all future readings, the corresponding offset values were subtracted from the raw accelerometer and gyroscope data. The corrected IMU readings were used in subsequent sensor calculations. The drift of the IMU is estimated and corrected using zero velocity updates (ZVU) [24], which are implemented during periods when the IMU is stationary. The basic idea behind ZVU is that when the IMU is not moving, its velocity and acceleration should ideally be zero. Fig. 10 shows the IMU X, Y, and Z axes readings after applying both the drift and offset correction of the accelerometer. Similar corrections are applied to the gyroscope. As anticipated, the performance improved, and the results are tabulated in Table IV.

H. Right and left ear data

We separately collected acoustic data from both the left and right ears from a subgroup of 18 participants (11 males and 7 females). Our analysis revealed variations in the responses between the two ears. The averaged PCA-transformed data for each subject remains distinct, demonstrating the uniqueness of each individual's ear-canal response.

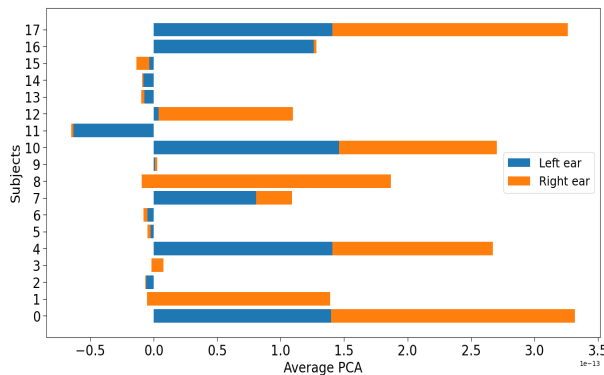


Fig. 11: Mean PCA data for the combined left and right ears

We separately collected acoustic data from both the left and right ears from a subgroup of 18 participants (11 males and 7 females). Our analysis revealed variations in the responses between the two ears. The averaged PCA-transformed data for each subject remains distinct, demonstrating the uniqueness of each individual's ear-canal response. This phenomenon is similar to fingerprints, where each finger of an individual has its own unique pattern. Once the data for both the left and right ears are enrolled, subsequent authentication can be performed using either ear, or the combined data can be used for authentication.

To determine the most effective strategy for storing left and right ear data for future authentication, we further explored multiple approaches, such as adding, subtracting, or averaging the features from both ears, but found these methods led to sparse and less robust representations due to the nature of data, as shown in Fig. 11. In contrast, concatenating the left

and right ear features preserved the integrity of the biometric data and significantly improved authentication performance.

Each participant was enrolled using concatenated data from both ears to build the authentication model. During verification, newly collected data was also concatenated and compared with the stored profile. The matching accuracy when both the ear data are used is 98.6%. This is comparable to using data from either of the individual ear. Thus, concatenation can be a reliable approach to capture the comprehensive biometric identity of an individual.

I. Fingerprint and ear-canal shape-based authentication

We conducted a comparative evaluation of fingerprint and ear-acoustic authentication data collected from the same 30 participants mentioned previously. Fingerprint recognition, a well-established and mature biometric modality, was used as the ground truth for validation. Each subject's ear-canal acoustic response was captured. The acoustic signatures, extracted from the reflected audio signals, were processed and matched against the enrolled templates. The resulting authentication outcomes were validated against the fingerprint-based identities. Our system achieved a matching accuracy of 100% across all participants, demonstrating the reliability and effectiveness of ear-based acoustic authentication as a biometric modality.

VI. CONCLUSIONS

In this work, we designed and implemented a novel ubiquitous two-factor authentication (2FA) system for IoT applications, integrating fingerprint recognition with ear-canal shape-based acoustic authentication. The capacitive fingerprint addresses the challenges posed by wet or oily fingers, providing consistent and accurate verification. To enhance the system's robustness, we introduced an additional modality based on the acoustic characteristics of the human ear-canal. This involves transmitting an audio signal through the earphones and capturing the reflected echoes using the integrated microphone to derive a unique biometric signature. The IMU sensor was embedded near the microphone and speaker units to accurately capture the positional variations in earphone placement. Experimental evaluation conducted on 30 human subjects demonstrated a recall of 100% matching for earphone position variations up to $\pm 25^\circ$ with a very low EER of 0.7%. The offset and drift corrections are applied to the IMU sensor to maintain consistent and reliable data collection over prolonged usage of the sensor, the system exhibited an improved recall over a one-month period. The ubiquitous earbud as an authentication device is an interesting approach.

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